

Approximate Determination of Rovibrational Densities of States $\rho(E, J)$ and Numbers of States $W(E, J)$

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Compact approximate equations for rovibrational densities of states $\rho(E, J)$ for harmonic oscillator-rigid rotor molecular systems are derived. Linear molecules, spherical tops, prolate and oblate symmetric tops are considered. Vibrational densities of states are represented by the Whitten-Rabinovitch approximation which then is convoluted with the contribution from the K -rotor (for a symmetric top). Likewise, numbers of states $W(E, J)$ are approximated for prolate symmetric top molecules. The derived expressions allow for quick calculations of specific rate constants $k(E, J)$ of unimolecular reactions.

1. Introduction

The statistical theory of unimolecular reactions expresses the specific rate constant $k(E, J)$ for reactions of molecules with a given total energy E and total angular momentum (quantum number J) by

$$k(E, J) = W^\ddagger(E, J) / h \rho(E, J) \quad (1.1)$$

In the framework of the statistical adiabatic channel model (SACM) [1], $W^\ddagger(E, J)$ denotes the number of open reaction channels for a given E and J ; in RRKM theory, $W^\ddagger(E, J)$ corresponds to the number of activated complex states with an energy below E and at fixed J [2]. In phase space theory (PST) or loose activated complex theory, $W^\ddagger(E, J)$ is easily calculated for low J and high J (see Appendix C of [3]); suitable low J -high J interpolations have also been derived [4], facilitating the calculation of $W^\ddagger(E, J)$ in the loose activated complex limit. In reality, the anisotropy of the potential surface introduces rigidity such that $W^\ddagger(E, J)$ falls below $W^\ddagger(E, J)_{\text{PST}}$ [5].

Besides $W^\ddagger(E, J)$, Eq. (1.1) contains the rovibrational density of states $\rho(E, J)$ of the reacting molecule. In traditional unimolecular rate theory, the J -dependence of $\rho(E, J)$ is often neglected and $\rho(E, J=0)$ is used instead; furthermore, anharmonicity is omitted such that simple state counting procedures are applicable such as the Beyer-Swinehart algorithm [6, 7] or the surprisingly successful empirical Whitten-Rabinovitch approximation [8, 9]. The anharmonicity question still represents a largely unsolved problem which, however, is attracting increasing attention. On the basis of real potential surfaces and/or experimentally determined accurate level positions, realistic anharmonicity corrections to the harmonic $\rho(E, J)$ are derived for energies below the reaction thresholds $E_0(J)$, see e.g. [3, 10–14]. Accounting for Morse anharmonicities and stretch-bend couplings, approximate anharmonicity factors

$$F_{\text{anh}}(E, J=0) = \rho(E, J=0) / \rho(E, J=0)_{\text{harm}} \quad (1.2)$$

of the type [10]

$$F_{\text{anh}}(E, J=0) \approx \prod_{j=1}^m (1 + E/(2s-3)D_j)^2 \quad (1.3)$$

tentatively have been proposed for $E \leq E_0$. (D_j denotes the Morse dissociation energies of the m stretching modes of the molecule.) For $E \geq E_0$, an increase of F_{anh} beyond Eq. (1.3) is obtained from the locations of the activated complexes such as described by SACM [10], because the molecular phase space in the dissociative continuum is limited by the positions of individual adiabatic channel maxima.

Leaving aside the difficult problems of determining rigidity factors in $W^\ddagger(E, J)$ and anharmonicity factors in $\rho(E, J)$, which both are related to the real potential energy surface, one is left with the much simpler task of calculating rigid rotor-harmonic oscillator rovibrational densities of states $\rho(E, J)_{\text{harm}}$. Approaching this problem, at first, one assumes that the kinetically relevant $\rho(E, J)$ corresponds to molecules, for which total E and total J are fixed while individual vibrational as well as rotational quantum numbers (such as K for a symmetric top) are not conserved on the time scale of the considered process. Then all quantum states compatible with a given E and J are included in the harmonic number of rovibrational states $W(E, J)_{\text{harm}}$ and the corresponding

$$\rho(E, J)_{\text{harm}} = \Delta W(E, J)_{\text{harm}} / \Delta E \quad (1.4)$$

The present article elaborates simplified expressions for $\rho(E, J)_{\text{harm}}$ (the index harm is omitted in the following). Although $W(E, J)$ can be obtained by state counting, when explicit expressions of all energy levels are given, there is a large practical interest to have a quicker approach, because the molecules can have many oscillators and many J may be of interest. For this reason, we discuss how in the case of linear, spherical top, oblate and prolate symmetric top mol-

ecules, rotational contributions can be combined with the Whitten-Rabinovitch approximation for vibrational densities of states. Our work continues earlier attempts from Appendix C of [3] and [15] as well as recent suggestions from [11]. The performance of the recommended approximate expressions, for a series of idealized molecular systems, is demonstrated by comparison with accurately counted state densities. The need for economic ways to calculate $W^\#(E, J)$ and $\rho(E, J)$ is exemplified by the sparsity of explicit calculations of $k(E, J)$ in the literature (see the systems treated in [1, 16–19]).

One may ask why, after the formulation of the elegant Beyer-Swinehart algorithm [6, 7] and the Whitten-Rabinovitch approximation [8, 9], there is still a need for approximations of the type discussed in the present work. Unfortunately, for real molecular systems with anharmonic, coupled oscillators as well as strong rovibrational couplings, neither of the two quick approaches apply. These complications are even not at all understood quantitatively for excited polyatomic molecules [10]. In this situation, it is often useful to combine simplified analytical anharmonicity and coupling models with analytical expressions of the here described type. The present work for this purpose provides practical tools by which state density calculations can be extended beyond the unfortunately not sufficiently realistic Beyer-Swinehart, Whitten-Rabinovitch levels.

2. Rovibrational Densities of States $\rho(E, J)$

In the rigid rotor-harmonic oscillator model considered here, a certain part $E_{\text{rot}}(J)$ of the total energy E is assumed to be fixed, while, for symmetric tops, another part $E_{\text{rot}}(K)$ can be varied without changing E and J . Therefore, the total rovibrational density of states follows by convolution of the internal density of states ρ_{int} (for s vibrations and r internal rotations) with that of the K -rotor. The corresponding expression is

$$\rho(E, J) = \sum_K' g(K) \rho_{\text{int}}(E - E_{\text{rot}}(K) - E_{\text{rot}}(J)) , \quad (2.1)$$

where the sum $\sum_K'$ includes those K in the range $-J \leq K \leq J$, for which $E - E_{\text{rot}}(K) - E_{\text{rot}}(J) \geq 0$, and where $g(K) = 2$ except for $g(K = 0) = 1$. The $(2J + 1)$ -fold M -degeneracy as well as nuclear spin statistical weights are omitted [1]. In order to arrive at simplified expressions, we replace the sum in Eq. (2.1) by an integral

$$\rho(E, J) \approx 2 \int_0^{J^*} \rho_{\text{int}}(E - E_{\text{rot}}(K) - E_{\text{rot}}(J)) dK , \quad (2.2)$$

where J^* is equal to J or is limited by the condition $E - E_{\text{rot}}(K) - E_{\text{rot}}(J) \geq 0$. Furthermore, we express $\rho_{\text{int}}(E')$ by the Whitten-Rabinovitch approximation [8, 9]

$$\rho_{\text{int}}(E') \approx \rho_{\text{WR}}(E') = \frac{\Gamma(1/2)^r (E' + a(E') E_z)^{s+r/2-1}}{\Gamma(s+r/2) \prod_{i=1}^s \varepsilon_i \prod_{j=1}^r B_j^{1/2}} \quad (2.3)$$

with

$$a(E') = 1 - \beta w(E') , \quad (2.4)$$

$$\beta = \frac{(s-1)(s+r/2)}{s} \frac{\sum \varepsilon_i^2}{(\sum \varepsilon_i)^2} , \quad (2.5)$$

and

$$\lg w = -1.0506(E'/E_z)^{0.25} \quad \text{at } E'/E_z \geq 1 \quad (2.6)$$

or

$$w^{-1} = 5(E'/E_z) + 2.73(E'/E_z)^{0.5} + 3.51 \quad (2.7)$$

at $0.1 \leq E'/E_z \leq 1$,

where r free internal rotors of rotational constants B_j and s harmonic oscillators of quanta $h\nu_i = \varepsilon_i$ are considered and the zeropoint energy $E_z = \sum \varepsilon_i/2$; Γ denotes the gamma function; symmetry numbers are omitted. The analytical convolution of Eq. (2.2) with Eq. (2.3) will be described in the following. The result is expressed in the form [3]

$$\rho(E, J) \approx \rho_{\text{WR}}(E^*) F_\rho(E^*, J) \quad (2.8)$$

with $E^* = E - BJ(J+1)$ for spherical and prolate symmetric tops (different values for oblate tops, see below). One could improve the result, if the “rotational degeneracy factor” $F_\rho(E^*, J)$ is not derived with the Whitten-Rabinovitch approximation but $\rho_{\text{WR}}(E^*)$ in Eq. (2.8) is replaced by a Beyer-Swinehart counted vibrational density of states.

2.1 Spherical Top

For a spherical top with rotational constants $A \approx B \approx C$, the calculation of the rotational degeneracy factor $F_\rho(E^*, J)$ is trivial [3]. With

$$E^* = E - BJ(J+1) \quad (2.9)$$

one obtains

$$F_\rho(E^*, J) = 2J + 1 . \quad (2.10)$$

2.2 Prolate Symmetric Top

For a prolate symmetric top molecule with the rotational constants $A > B = C$, Eq. (2.2) becomes

$$\rho(E, J) \approx 2 \int_0^{J^*} \rho_{\text{int}}(E - BJ(J+1) - (A-B)K^2) dK . \quad (2.11)$$

Obviously one has $E \geq BJ(J+1)$, because E includes the rotational energy. For a given combination of E and J , the upper limit of the integral is fixed by the condition

$$J^* = \min \left(J, \sqrt{\frac{E - BJ(J+1)}{A - B}} \right). \quad (2.12)$$

In the following, for notational simplicity, we set $r = 0$. If internal rotors have to be included, in all expressions, one simply has to replace s by $s + r/2$. Eq. (2.11) with ρ_{int} from Eq. (2.3) can be solved analytically and yields

$$\rho(E, J) \approx \rho_{\text{WR}}(E^*) \left(\frac{E^* + a(E^*)E_z}{A - B} \right)^{1/2} \times \frac{\Gamma(1/2)\Gamma(s)}{\Gamma(s+1/2)} I_x(1/2, s) \quad (2.13)$$

with $E^* = E - BJ(J+1)$ and

$$x = \frac{(A - B)J^2}{E^* + a(E^*)E_z}. \quad (2.14)$$

In deriving Eq. (2.13), E^* was replaced by $E^* + a(E^*)E_z$, and the approximation $a(E^* - (A - B)K^2) \approx a(E^*)$ was used. I_x denotes the regularized incomplete beta function [20], which varies between zero and unity as x goes from 0 to 1. If the r.h.s. of Eq. (2.14) is larger than unity (the high- J case), one has to set $x = 1$, because then the complete beta function will appear in Eq. (2.13). In this case, the external rotation contributes to the density of states just as a one dimensional free rotor with an effective rotational constant $(A - B)$. In the low- J case, however, for decreasing J , this contribution is successively “frozen”, indicated by a decreasing $I_x < 1$. To prevent Eq. (2.13) from going to zero for $J \rightarrow 0$, one has to add unity to F_ρ . We try to represent $I_x(1/2, s)$ by an empirical formula. We found that the following approximation gives very good results:

$$I_x(1/2, s) \approx \tanh \left[\frac{\Gamma(s+1/2)}{\Gamma(s)\Gamma(1/2)} \left(1 + \frac{2+s}{10} x \right) \frac{x^{1/2}}{0.5} \right]. \quad (2.15)$$

The largest deviation between the exact $I_x(1/2, s)$ and Eq. (2.15) over the whole range $0 \leq x \leq 1$ is $\approx 2.1\%$ for $s = 3$, $\approx 1.4\%$ for $s = 4$, and $< 1\%$ for $s > 4$. In most cases the deviations are much smaller than 1% even for $s = 3$ and 4 . Eq. (2.15), in addition, has the advantage that the high- J and low- J case are covered simultaneously because it very well represents the correct limiting behaviour of Eq. (2.13), namely $I_x = 1$ for $x \geq 1$. Combining Eqs. (2.13) and (2.15), one finally obtains

$$F_\rho \approx \left(\frac{J^2}{x} \right)^{1/2} \frac{\Gamma(1/2)\Gamma(s)}{\Gamma(s+1/2)} \times \left\{ \tanh \left[\frac{\Gamma(s+1/2)}{\Gamma(s)\Gamma(1/2)} \left(1 + \frac{2+s}{10} x \right) \frac{x^{1/2}}{0.5} \right] \right\} + 1, \quad (2.16)$$

where the factor $(J^2/x)^{1/2}$, is used as an abbreviation for $[(E^* + aE_z)/(A - B)]^{1/2}$, see Eq. (2.14).

Results for a number of molecules are shown in Figs. 1–3 and Table 1. The accuracy of our results is not worse than that of the Whitten-Rabinovitch approximation

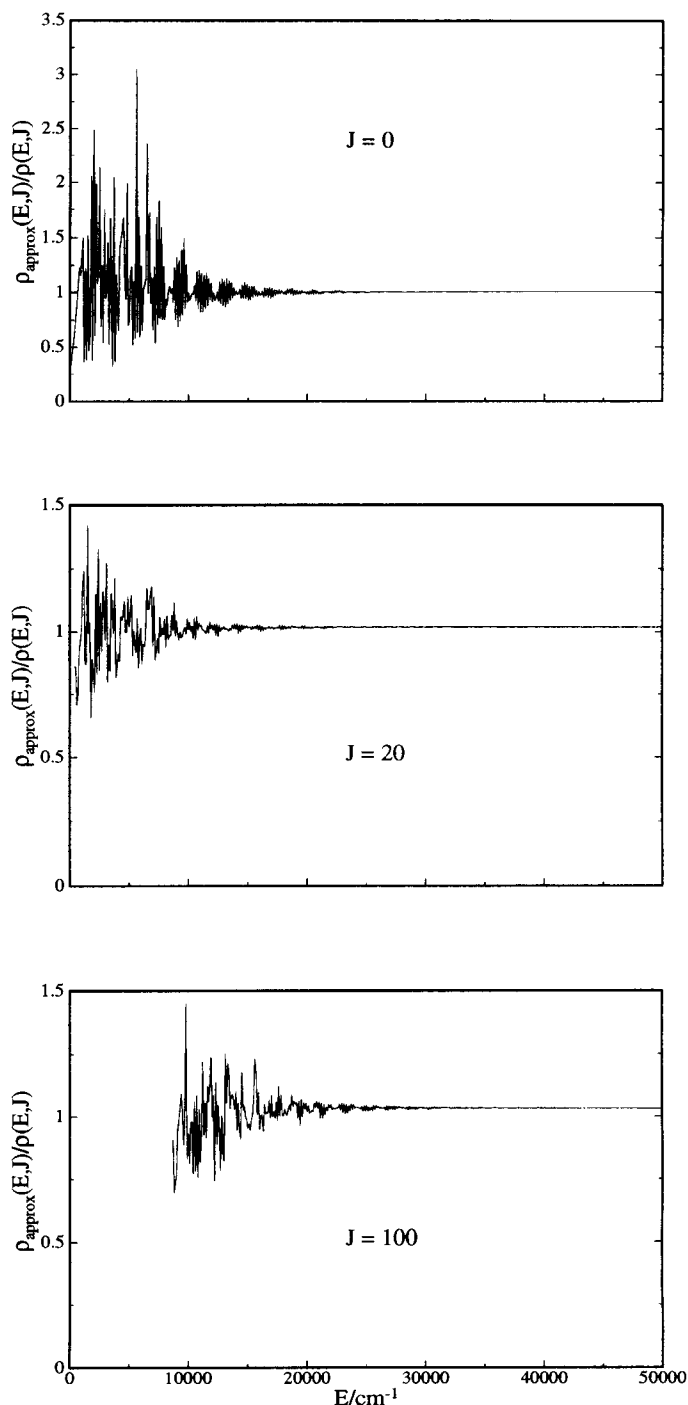


Fig. 1
Ratio of the approximate and the exactly counted density of states for H_2O_2 . The curves start at $E = BJ(J+1)$, the case $J = 0$ also represents the Whitten-Rabinovitch approximation alone. The irregular variations at low energies are due to the discontinuous nature of the exactly counted density of states

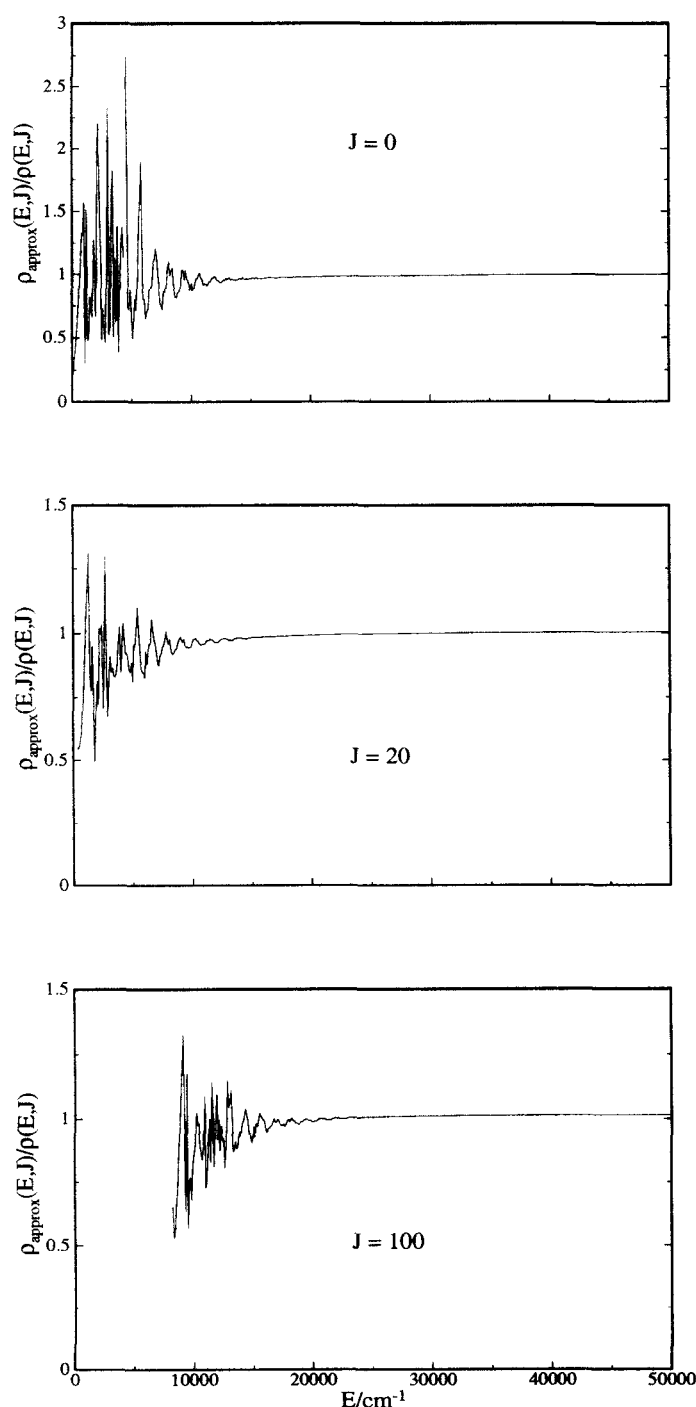


Fig. 2
As Fig. 1, but for CH_3OH

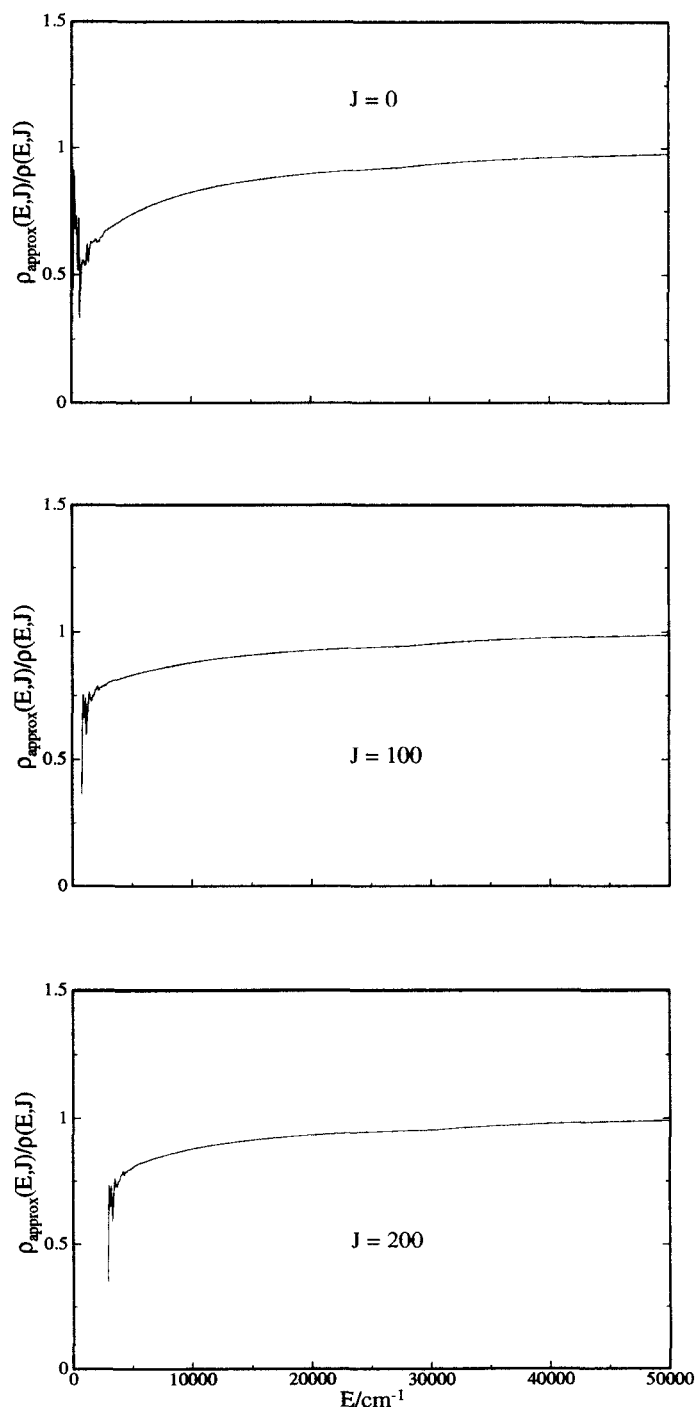


Fig. 3
As Fig. 1, but for C_7H_8

represented by $J = 0$. In the case of larger molecules, the latter seems to underestimate somewhat the state density at lower energies as can be recognized especially in Fig. 3. This situation can be improved by the use of a redefined β following [21] instead of Eq. (2.5). However, in order to retain the simplicity of the model and because densities of states are normally needed at higher energies, we do not use this modification here.

2.3 Oblate Symmetric Top

An oblate symmetric top molecule is characterized by the rotational constants $A = B > C$. In this case, Eq. (2.2) leads to

$$\rho(E, J) \approx 2 \int_{K^*}^J \rho_{\text{int}}(E - BJ(J+1) + (B-C)K^2) dK \quad (2.17)$$

Table 1

Comparison of exactly counted densities of states $\rho(E, J)$ with approximate values $\rho_{\text{approx}}(E, J)$ from Eqs. (2.3)–(2.7), (2.13), (2.14), and (2.16) for prolate symmetric top molecules molecular parameters: a) for H_2O_2 [18], b) for CH_3OH [23], c) for C_2H_6 [1], d) for C_3H_8 [24], e) for C_7H_8 [25])

E/cm^{-1}	J	$\rho(E, J)/\text{cm}$	$\rho_{\text{approx}}(E, J)/\text{cm}$
10000	$\text{H}_2\text{O}_2^{\text{a}}$	$(E_z = 5454 \text{ cm}^{-1})$	
	0	$9.50 \cdot 10^{-1}$	$9.44 \cdot 10^{-1}$
	5	$9.35 \cdot 10^0$	$1.01 \cdot 10^1$
	10	$1.83 \cdot 10^1$	$1.74 \cdot 10^1$
	20	$2.34 \cdot 10^1$	$2.36 \cdot 10^1$
	50	$1.26 \cdot 10^1$	$1.29 \cdot 10^1$
40000	100	$1.49 \cdot 10^{-1}$	$1.39 \cdot 10^{-1}$
	0	$2.33 \cdot 10^2$	$2.35 \cdot 10^2$
	5	$2.53 \cdot 10^3$	$2.56 \cdot 10^3$
	10	$4.67 \cdot 10^3$	$4.73 \cdot 10^3$
	20	$7.98 \cdot 10^3$	$8.12 \cdot 10^3$
	50	$9.16 \cdot 10^3$	$9.38 \cdot 10^3$
20000	100	$3.74 \cdot 10^3$	$3.86 \cdot 10^3$
	$\text{CH}_3\text{OH}^{\text{b}}$	$(E = 10868 \text{ cm}^{-1})$	
	0	$3.48 \cdot 10^3$	$3.41 \cdot 10^3$
	5	$3.76 \cdot 10^4$	$3.68 \cdot 10^4$
	10	$6.76 \cdot 10^4$	$6.65 \cdot 10^4$
	20	$1.06 \cdot 10^5$	$1.05 \cdot 10^5$
50000	50	$7.18 \cdot 10^4$	$7.22 \cdot 10^4$
	100	$3.79 \cdot 10^3$	$3.75 \cdot 10^3$
	0	$7.38 \cdot 10^6$	$7.38 \cdot 10^6$
	5	$8.06 \cdot 10^7$	$8.04 \cdot 10^7$
	10	$1.49 \cdot 10^8$	$1.49 \cdot 10^8$
	20	$2.61 \cdot 10^8$	$2.63 \cdot 10^8$
20000	50	$3.22 \cdot 10^8$	$3.25 \cdot 10^8$
	100	$9.42 \cdot 10^7$	$9.56 \cdot 10^7$
	$\text{C}_2\text{H}_6^{\text{c}}$	$(E_z = 15873 \text{ cm}^{-1})$	
	0	$1.27 \cdot 10^4$	$1.23 \cdot 10^4$
	10	$2.47 \cdot 10^5$	$2.41 \cdot 10^5$
	50	$2.69 \cdot 10^5$	$2.69 \cdot 10^5$
50000	100	$1.24 \cdot 10^4$	$1.22 \cdot 10^4$
	150	$8.40 \cdot 10^0$	$8.26 \cdot 10^0$
	0	$6.22 \cdot 10^8$	$6.28 \cdot 10^8$
	10	$1.26 \cdot 10^{10}$	$1.27 \cdot 10^{10}$
	50	$2.71 \cdot 10^{10}$	$2.77 \cdot 10^{10}$
	100	$6.85 \cdot 10^9$	$7.03 \cdot 10^9$
20000	150	$4.41 \cdot 10^8$	$4.52 \cdot 10^8$
	200	$3.66 \cdot 10^6$	$3.73 \cdot 10^6$
	$\text{C}_3\text{H}_8^{\text{d}}$	$(E_z = 21862 \text{ cm}^{-1})$	
	0	$2.33 \cdot 10^8$	$2.17 \cdot 10^8$
	10	$4.73 \cdot 10^9$	$4.40 \cdot 10^9$
	50	$1.01 \cdot 10^{10}$	$9.63 \cdot 10^9$
50000	100	$2.42 \cdot 10^9$	$2.35 \cdot 10^9$
	200	$1.46 \cdot 10^6$	$1.38 \cdot 10^6$
	0	$8.82 \cdot 10^{14}$	$8.63 \cdot 10^{14}$
	10	$1.82 \cdot 10^{16}$	$1.78 \cdot 10^{16}$
	50	$5.60 \cdot 10^{16}$	$5.53 \cdot 10^{16}$
	100	$3.27 \cdot 10^{16}$	$3.25 \cdot 10^{16}$
20000	200	$1.09 \cdot 10^{15}$	$1.09 \cdot 10^{15}$
	$\text{C}_7\text{H}_8^{\text{e}}$	$(E_z = 27183 \text{ cm}^{-1})$	
	0	$2.51 \cdot 10^{12}$	$2.26 \cdot 10^{12}$
	50	$1.95 \cdot 10^{14}$	$1.77 \cdot 10^{14}$
	100	$1.83 \cdot 10^{14}$	$1.70 \cdot 10^{14}$
	200	$2.45 \cdot 10^{13}$	$2.29 \cdot 10^{13}$
50000	500	$2.74 \cdot 10^3$	$2.22 \cdot 10^3$
	0	$2.02 \cdot 10^{21}$	$1.98 \cdot 10^{21}$
	50	$1.77 \cdot 10^{23}$	$1.74 \cdot 10^{23}$
	100	$2.32 \cdot 10^{23}$	$2.29 \cdot 10^{23}$
	200	$9.93 \cdot 10^{22}$	$9.84 \cdot 10^{22}$
	500	$1.14 \cdot 10^{19}$	$1.10 \cdot 10^{19}$

with the general constraint $E \geq BJ(J+1) - (B-C)K^2$. Defining E^* by $E^* = E - BJ(J+1) + (B-C)J^2$, for the lower integration limit in the case $E \geq BJ(J+1)$ (case A), one obtains $K^* = 0$ and for $E < BJ(J+1)$ (case B), one obtains $K^* = [J^2 - E^*/(B-C)]^{1/2}$. Because the two cases lead to somewhat different results, we treat them separately, starting with case A. With

$$x = \frac{(B-C)J^2}{E^* + a(E^*)E_z} \quad (2.18)$$

Eq. (2.17) becomes

$$\rho(E, J) \approx \rho_{\text{WR}}(E^*) \left(\frac{E^* + a(E^*)E_z}{B-C} \right)^{1/2} \times \int_{1-x}^1 \frac{y^{s-1}}{(y+x-1)^{1/2}} dy, \quad (2.19)$$

where the integral formally can be expressed in terms of the hypergeometric function $F(\alpha, \beta, \gamma; z)$ [20]. It follows

$$F_{\rho, A} \approx \left(\frac{J^2}{x} \right)^{1/2} 2x^{1/2}(1-x)^{s-1} F\left(\frac{1}{2}, 1-s, \frac{3}{2}; \frac{x}{x-1}\right) + 1, \quad (2.20)$$

where unity is added again to prevent $\rho(E, J)$ from going to zero for $J = 0$. One should note that case A corresponds to $0 \leq x < 1$. In case B, i.e. $x \geq 1$, Eq. (2.17) gives

$$\rho(E, J) \approx \rho_{\text{WR}}(E^*) \left(\frac{E^* + a(E^*)E_z}{B-C} \right)^{1/2} \int_0^1 \frac{y^{s-1}}{(y+x-1)^{1/2}} dy, \quad (2.21)$$

which leads to

$$F_{\rho, B} \approx \left(\frac{J^2}{x} \right)^{1/2} s^{-1}(x-1)^{-1/2} F\left(\frac{1}{2}, s, s+1; \frac{1}{1-x}\right). \quad (2.22)$$

The hypergeometric function cannot generally be represented in a simple way. If s is an integer, however, a polynomial expansion of the degree $s-1$ is available for case A [20] and can be used in Eq. (2.20):

$$F\left(\frac{1}{2}, 1-s, \frac{3}{2}; \frac{x}{x-1}\right) = \sum_{i=0}^{s-1} \frac{\Gamma(s)}{(2i+1)\Gamma(i+1)\Gamma(s-i)} \left[\frac{x}{1-x} \right]^i. \quad (2.23)$$

In case B, the following approximation was found to give excellent results:

$$F_{\rho, B} \approx \left(\frac{J^2}{x} \right)^{1/2} x^{-1/2} \left[\frac{s}{s^2 - \frac{1}{4}} + \frac{1}{(2s^2 - \frac{1}{2})x} \right]. \quad (2.24)$$

The deviation between Eqs. (2.24) and (2.22) is generally less than 5% for $s > 3$, less than 1% for $s \geq 6$, and less than 0.1% for $s \geq 19$. We therefore recommend the use of Eq. (2.20) with Eq. (2.23) for $x \leq 1$ and Eq. (2.24) for $x \geq 1$. Both equa-

tions correctly converge towards the value of $F_\rho(x=1) = J/(s-1/2)$.

To demonstrate the quality of the approximation, two examples of different molecular complexity are given in Table 2. The accuracy is again comparable or even better than that of the Whitten-Rabinovitch method as can be recognized from Fig. 4.

Table 2

Comparison of exactly counted densities of states $\rho(E, J)$ with approximate values $\rho_{\text{approx}}(E, J)$ from Eqs. (2.3)–(2.7) and (2.18)–(2.24) for oblate symmetric top molecules molecular parameters: a) for CH_3 [1], b) for C_6H_6 [26])

E/cm^{-1}	J	$\rho(E, J)/\text{cm}$	$\rho_{\text{approx}}(E, J)/\text{cm}$
CH₃^{a)}			
$(E_z = 6370 \text{ cm}^{-1})$			
20000	0	$4.17 \cdot 10^{-1}$	$4.90 \cdot 10^{-1}$
	10	$6.51 \cdot 10^0$	$8.66 \cdot 10^0$
	20	$1.26 \cdot 10^1$	$1.01 \cdot 10^1$
	50	$1.77 \cdot 10^{-1}$	$1.97 \cdot 10^{-1}$
50000	0	$2.43 \cdot 10^1$	$2.27 \cdot 10^1$
	10	$4.43 \cdot 10^1$	$4.41 \cdot 10^1$
	20	$6.85 \cdot 10^2$	$6.86 \cdot 10^2$
	50	$2.65 \cdot 10^2$	$2.69 \cdot 10^2$
	70	$2.44 \cdot 10^1$	$2.61 \cdot 10^1$
C₆H₆^{b)}			
$(E_z = 21363 \text{ cm}^{-1})$			
30000	0	$4.61 \cdot 10^{11}$	$4.55 \cdot 10^{11}$
	30	$2.57 \cdot 10^{13}$	$2.54 \cdot 10^{13}$
	150	$1.55 \cdot 10^{13}$	$1.57 \cdot 10^{13}$
	400	$1.26 \cdot 10^8$	$1.27 \cdot 10^8$
60000	0	$5.75 \cdot 10^{17}$	$5.86 \cdot 10^{17}$
	30	$3.32 \cdot 10^{19}$	$3.39 \cdot 10^{19}$
	150	$4.64 \cdot 10^{19}$	$4.76 \cdot 10^{19}$
	400	$6.78 \cdot 10^{16}$	$7.05 \cdot 10^{16}$

2.4 Linear Triatomic Molecule

We now consider a linear triatomic molecule as a prototype for the influence of the vibrational angular momentum on the J -resolved density of states. The total energy may be written as [22]

$$E = \varepsilon_1(v_1 + 1/2) + \varepsilon_2(v_2 + 1/2) + \varepsilon_b(v'_b + 1) + BJ(J+1) - Bl^2 \quad (2.25)$$

with the vibrational quantum numbers v_i and the vibrational angular momentum quantum number l . The subscripts 1 and 2 refer to the stretching and the subscript b to the degenerate bending vibrations 3 and 4 with $\varepsilon_b = \varepsilon_3 = \varepsilon_4$ and $v'_b = v_3 + v_4$. For l , the conditions $l = v'_b, v'_b - 2, v'_b - 4, \dots, -v'_b$ and $|l| \leq J$ apply [22]. The classical density of states of the two stretching vibrations is $\rho_{12}(E) = E/(\varepsilon_1 \varepsilon_2)$ and that of the bending vibration $\rho_b(E, l) = 1/(2\varepsilon_b)$. The factor $1/2$ in the latter case results from the first condition for l mentioned above. Integration over all accessible l leads to $\rho_b(E, J) = J/\varepsilon_b$ for $J \leq E/\varepsilon_b$ and $\rho_b(E) = E/\varepsilon_b^2$ otherwise. Neglecting, as usual, the M -degeneracy, the combined density of states of all oscillators follows by the convolution

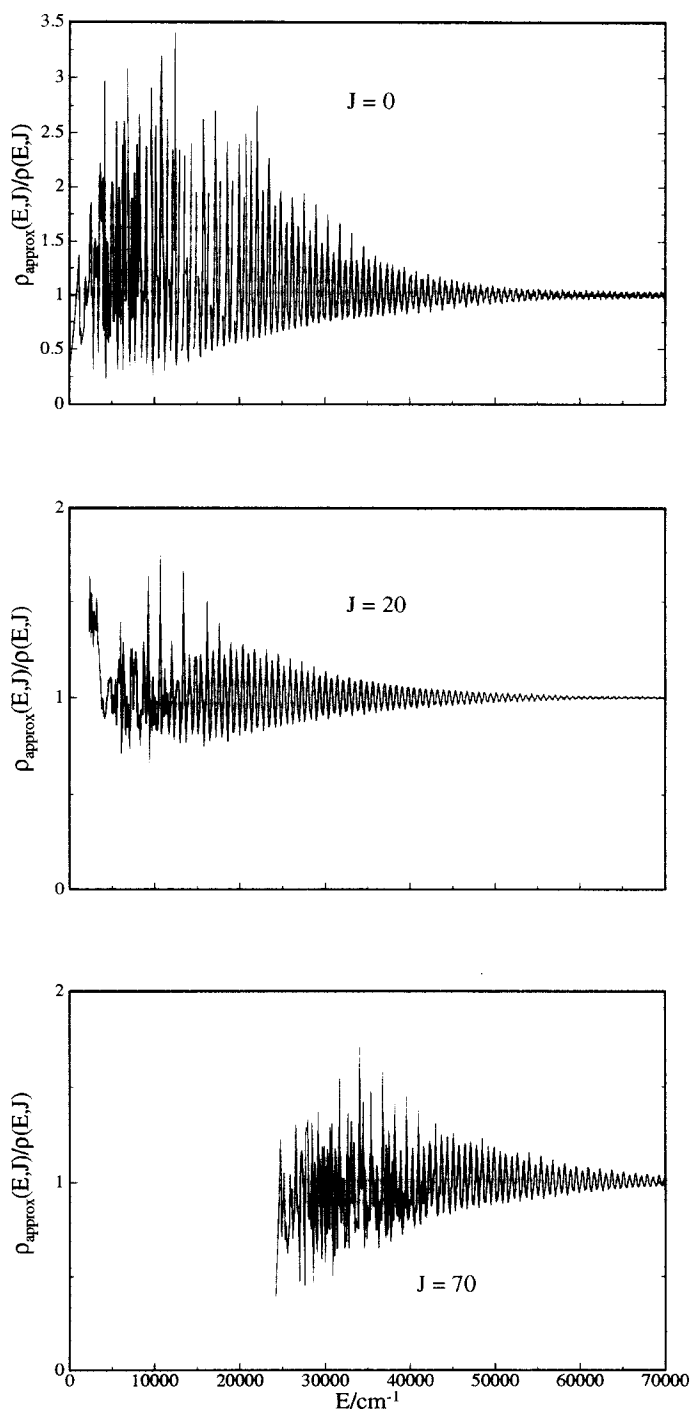


Fig. 4
As Fig. 1, but for CH_3

$$\rho(E, J) \approx \int_0^{E^*} \int_{**} \rho_{12}(E^* - y + Bl^2) \rho_b(y, l) dl dy \quad (2.26)$$

where $**$ indicates that l varies between $-v'_b$ and v'_b for $0 \leq y \leq J\varepsilon_b$ and between $-J$ and J for $J\varepsilon_b \leq y \leq E^*$ with $E^* = E - BJ(J+1)$. One should note that the argument of ρ_{12} is increased by the term Bl^2 which, according to Eq. (2.25), represents an energy flow from the external

rotation into the internal degrees of freedom with increasing l . Substituting E^* by $E^* + aE_z$, J by $J + 1$, and defining

$$x = \frac{(J+1)\varepsilon_b}{E^* + a(E^*)E_z} \quad (2.27)$$

from Eq. (2.26) one obtains

$$\rho(E, J) \approx \rho_{\text{int}}(E^*) \left[3x + \left(\frac{2B(J+1)}{\varepsilon_b} - 3 \right) x^2 + \left(1 - \frac{3B(J+1)}{2\varepsilon_b} \right) x^3 \right] \quad (2.28)$$

for $x \leq 1$ and

$$\rho(E, J) \approx \rho_{\text{int}}(E^*) \left[1 + \frac{B(J+1)}{2\varepsilon_b x} \right] \quad (2.29)$$

for $x \geq 1$, where ρ_{int} refers to all four oscillators. Using the Whitten-Rabinovitch approximation for $\rho_{\text{int}}(E^*)$, one obtains in the case of low J

$$\rho(E, \text{low } J) \approx (J+1) \frac{(E^* + a(E^*)E_z)^2}{\Gamma(3) \prod_{i=1}^3 \varepsilon_i} \quad (2.30)$$

i.e. $(J+1)$ times the density of states for an ensemble of three oscillators. In the case of high J , Eq. (2.29) applies. Thus, the unimolecular rate constant $k(E, J)$ is overestimated for high J and underestimated for low J , if the vibrational angular momentum coupling is neglected. Moreover, even in the high- J case, where the density of states becomes independent of J , an energy dependent coupling factor, larger than unity appears (because of Eq. (2.27), Eq. (2.29) is independent of J).

Exactly counted and semiclassical estimated values are compared in Table 3, and the influence of the energy on the accuracy of the approximation is demonstrated in Fig. 5.

3. Rovibrational Sums of States $W(E, J)$ for Prolate Symmetric Top Molecules

The analytical treatment of Section 2 can easily be extended to sums of states $W(E, J)$. We demonstrate this for the case of prolate symmetric top molecules.

In complete analogy to Eq. (2.11), $W(E, J)$ is given by

$$W(E, J) \approx 2 \int_0^{J^*} W_{\text{int}}(E - BJ(J+1) - (A-B)K^2) dK \quad (3.1)$$

with J^* from Eq. (2.12). Using the Whitten-Rabinovitch approximation for the sums of states

$$W_{\text{int}}(E') \approx W_{\text{WR}}(E') = \frac{\Gamma(1/2)^r (E' + a(E')E_z)^{s+r/2}}{\Gamma(s+1+r/2) \prod_{i=1}^s \varepsilon_i \prod_{j=1}^r B_j^{1/2}}, \quad (3.2)$$

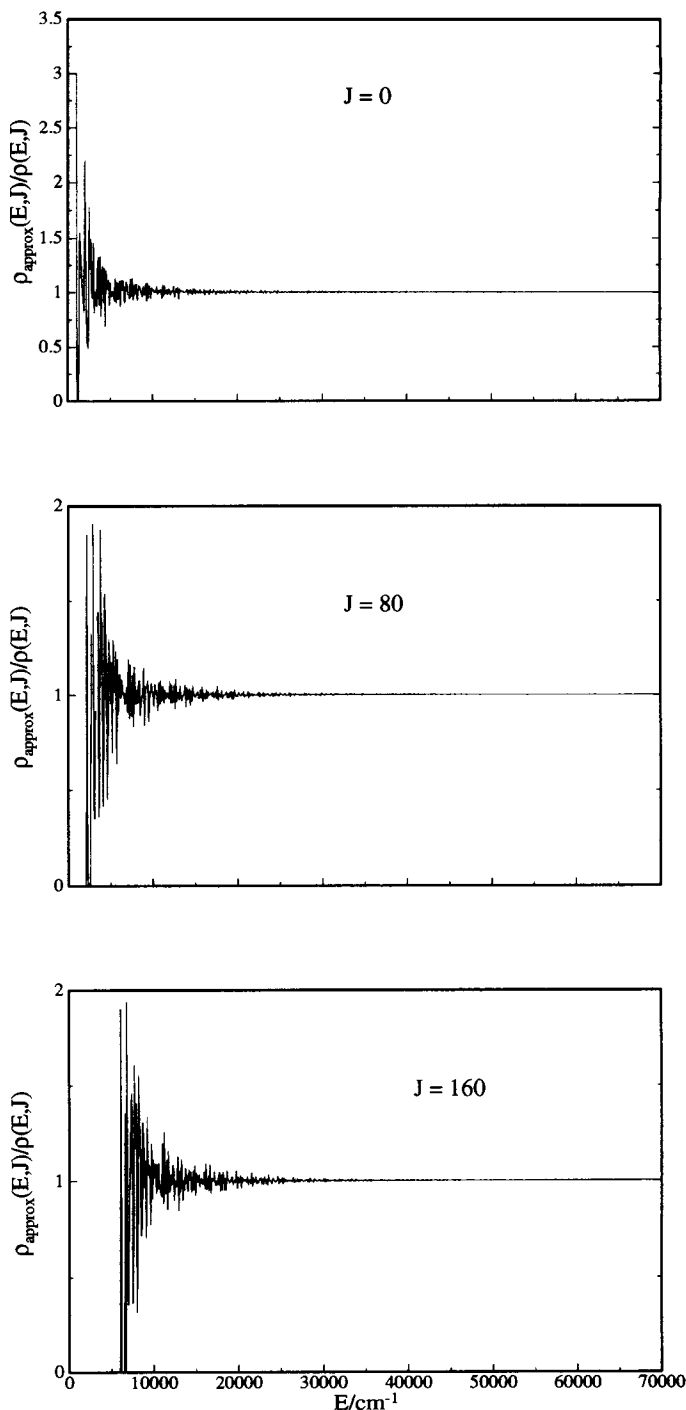


Fig. 5
As Fig. 1, but for OCS

(see Eqs. (2.4) – (2.7)), one obtains by analytical integration

$$W(E, J) \approx W_{\text{WR}}(E^*) \left(\frac{E^* + a(E^*)E_z}{A-B} \right)^{1/2} \times \frac{\Gamma(1/2)\Gamma(s+1)}{\Gamma(s+3/2)} I_x(1/2, s+1) . \quad (3.3)$$

Table 3

Comparison of exactly counted densities of states $\rho(E, J)$ with approximated values $\rho_{\text{approx}}(E, J)$ from Eqs. (2.3)–(2.7), (2.28), and (2.29) for linear triatomic molecules molecular parameters: a) for HCN [27], b) for OCS [2]

E/cm^{-1}	J	$\rho(E, J)/\text{cm}$	$\rho_{\text{approx}}(E, J)/\text{cm}$
HCN^{a)} ($E_z = 3418 \text{ cm}^{-1}$)			
30 000	0	$1.11 \cdot 10^{-1}$	$1.10 \cdot 10^{-1}$
	30	$1.55 \cdot 10^0$	$1.54 \cdot 10^0$
	60	$1.07 \cdot 10^0$	$1.07 \cdot 10^0$
	90	$4.61 \cdot 10^{-1}$	$4.63 \cdot 10^{-1}$
	120	$7.25 \cdot 10^{-2}$	$7.82 \cdot 10^{-2}$
70 000	0	$5.39 \cdot 10^{-1}$	$5.38 \cdot 10^{-1}$
	30	$1.19 \cdot 10^1$	$1.19 \cdot 10^1$
	60	$1.49 \cdot 10^1$	$1.49 \cdot 10^1$
	90	$1.21 \cdot 10^1$	$1.18 \cdot 10^1$
	120	$7.21 \cdot 10^0$	$7.07 \cdot 10^0$
	150	$3.19 \cdot 10^0$	$3.16 \cdot 10^0$
OCS^{b)} ($E_z = 1980 \text{ cm}^{-1}$)			
30 000	0	$5.51 \cdot 10^{-1}$	$4.45 \cdot 10^{-1}$
	40	$1.08 \cdot 10^1$	$1.07 \cdot 10^1$
	80	$1.02 \cdot 10^0$	$1.01 \cdot 10^0$
	120	$8.61 \cdot 10^0$	$8.59 \cdot 10^0$
	160	$6.79 \cdot 10^0$	$6.71 \cdot 10^0$
	200	$4.75 \cdot 10^0$	$4.73 \cdot 10^0$
70 000	0	$2.80 \cdot 10^0$	$2.79 \cdot 10^0$
	40	$8.39 \cdot 10^1$	$8.38 \cdot 10^1$
	80	$1.16 \cdot 10^2$	$1.16 \cdot 10^2$
	120	$1.18 \cdot 10^2$	$1.17 \cdot 10^2$
	160	$1.06 \cdot 10^2$	$1.06 \cdot 10^2$
	200	$9.29 \cdot 10^1$	$9.26 \cdot 10^1$

Here again, $E^* = E - BJ(J+1)$ and, for convenience, r was set equal to zero (see remarks following Eq. (2.12)). If $I_x(1/2, s+1)$ is represented by Eq. (2.15), one can express Eq. (3.3) by

$$W(E, J) = W_{\text{WR}}(E^*) F_W(E^*, J) \quad (3.4)$$

with

$$F_W \approx \left(\frac{J^2}{x} \right)^{1/2} \frac{\Gamma(1/2) \Gamma(s+1)}{\Gamma(s+3/2)} \times \left\{ \tanh \left[\frac{\Gamma(s+3/2)}{\Gamma(s+1) \Gamma(1/2)} \left(1 + \frac{3+s}{10} x \right) \frac{x^{1/2}}{0.5} \right] \right\} + 1, \quad (3.5)$$

where x is defined by Eq. (2.14).

Results for three different prolate symmetric top molecules are shown in Table 4, and the energy dependence is demonstrated in Figs. 6–8. It is again to be seen that the deviations are much smaller than 10% in most cases except at very low energies.

Table 4

Comparison of exactly counted sums of states $W(E, J)$ with approximate values $W_{\text{approx}}(E, J)$ from Eqs. (3.2) and (3.4)–(3.6) for prolate symmetric top molecules (for molecular parameters see Table 1)

E/cm^{-1}	J	$W(E, J)$	$W_{\text{approx}}(E, J)$
H₂O₂			
5000	0	$1.81 \cdot 10^2$	$1.82 \cdot 10^2$
	5	$1.83 \cdot 10^3$	$1.89 \cdot 10^3$
	10	$2.89 \cdot 10^3$	$2.92 \cdot 10^3$
	20	$2.94 \cdot 10^3$	$3.09 \cdot 10^3$
	50	$6.02 \cdot 10^3$	$6.46 \cdot 10^3$
10000	0	$2.35 \cdot 10^3$	$2.37 \cdot 10^3$
	5	$2.45 \cdot 10^4$	$2.51 \cdot 10^4$
	10	$4.15 \cdot 10^4$	$4.21 \cdot 10^4$
	20	$5.20 \cdot 10^4$	$5.37 \cdot 10^4$
	50	$2.19 \cdot 10^4$	$2.30 \cdot 10^4$
	100	$9.10 \cdot 10^1$	$1.09 \cdot 10^2$
CH₃OH			
5000	0	$1.03 \cdot 10^3$	$1.02 \cdot 10^3$
	5	$1.07 \cdot 10^4$	$1.07 \cdot 10^4$
	10	$1.77 \cdot 10^4$	$1.80 \cdot 10^4$
	20	$2.02 \cdot 10^4$	$2.00 \cdot 10^4$
	50	$2.83 \cdot 10^4$	$2.92 \cdot 10^4$
10000	0	$5.20 \cdot 10^4$	$5.16 \cdot 10^4$
	5	$5.52 \cdot 10^5$	$5.51 \cdot 10^5$
	10	$9.50 \cdot 10^5$	$9.59 \cdot 10^5$
	20	$1.28 \cdot 10^6$	$1.27 \cdot 10^6$
	50	$4.17 \cdot 10^5$	$4.28 \cdot 10^5$
	100	$5.83 \cdot 10^2$	$5.81 \cdot 10^2$
C₇H₈			
5000	0	$3.97 \cdot 10^6$	$3.93 \cdot 10^6$
	50	$2.28 \cdot 10^8$	$2.39 \cdot 10^8$
	100	$9.37 \cdot 10^7$	$9.48 \cdot 10^7$
	200	$4.30 \cdot 10^5$	$4.67 \cdot 10^5$
	500	$2.00 \cdot 10^{10}$	$1.96 \cdot 10^{10}$
10000	0	$2.00 \cdot 10^{10}$	$1.96 \cdot 10^{10}$
	50	$1.35 \cdot 10^{12}$	$1.36 \cdot 10^{12}$
	100	$8.67 \cdot 10^{11}$	$8.48 \cdot 10^{11}$
	200	$2.92 \cdot 10^{10}$	$2.92 \cdot 10^{10}$
	300	$2.30 \cdot 10^7$	$2.41 \cdot 10^7$

4. Approximate Expressions for Specific Rate Constants $k(E, J)$

The derived expressions for harmonic rovibrational densities of states $\rho(E, J)_{\text{harm}}$ from Section 2 (together with the approximate anharmonicity factors from Eq. (1.3), their modifications for $J > 0$, and their additional contributions at $E > E_0$, see Section 1), facilitate the calculation of $\rho(E, J)$ in the specific rate constants $k(E, J)$ of Eq. (1.1). The determination of $W^{\ddagger}(E, J)$ for simple bond fission reactions on the level of phase space theory, by similar approximative techniques as applied in the present work, was described in [4]. Before final recommendations for $W^{\ddagger}(E, J)$ in this group of reactions can be made, E - and J -dependent rigidity factors have to be derived systematically. We, therefore, do not further discuss loose activated complex situations at this stage. Instead, we now consider rigid activated complex situations, where SACM and RRKM theory approach each other.

When the structural parameters of a rigid activated complex are known or estimated, the results of Section 3 can be

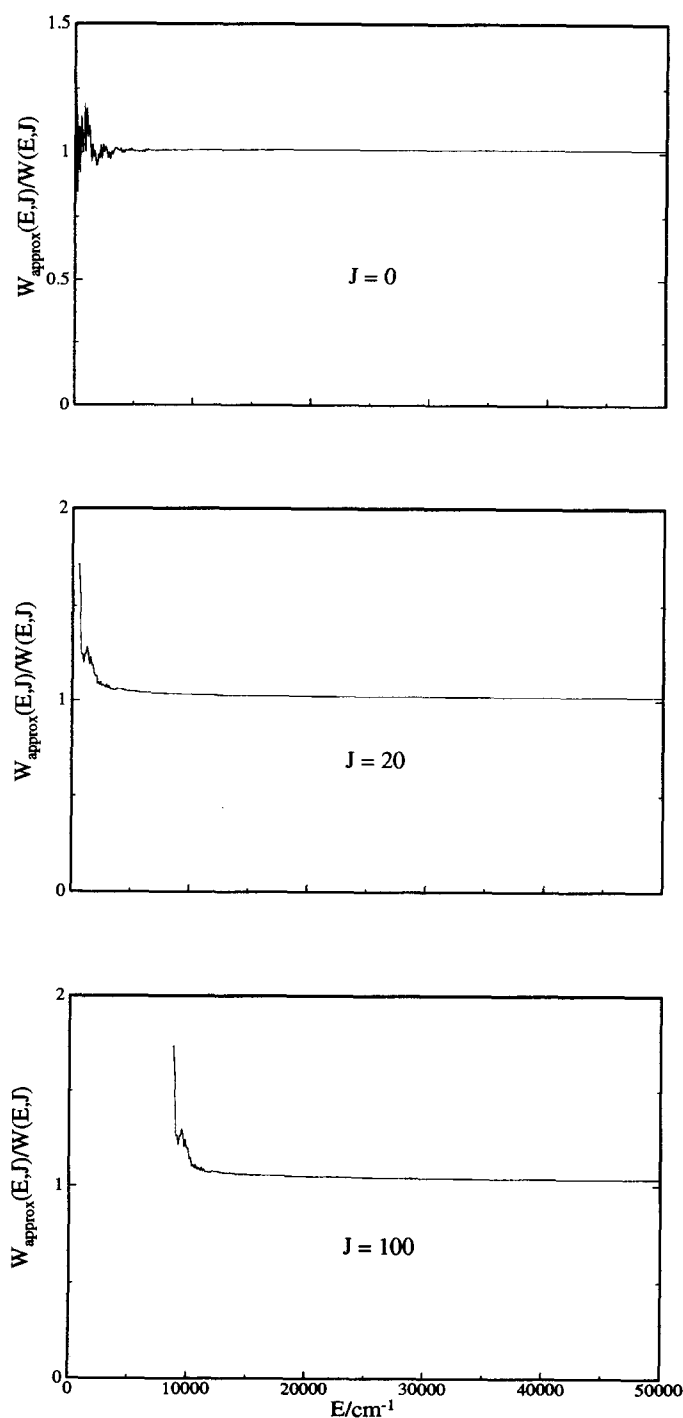


Fig. 6
Ratio of the approximate to the exactly counted sums of states for H_2O_2 . The curves start at $E = BJ(J+1)$, the case $J = 0$ also represents the Whitten-Rabinovitch approximation alone. The starting point can not be seen for $J = 20$ and $J = 100$, but the deviations become smaller than a factor of 2 within 200 cm^{-1} above $BJ(J+1)$

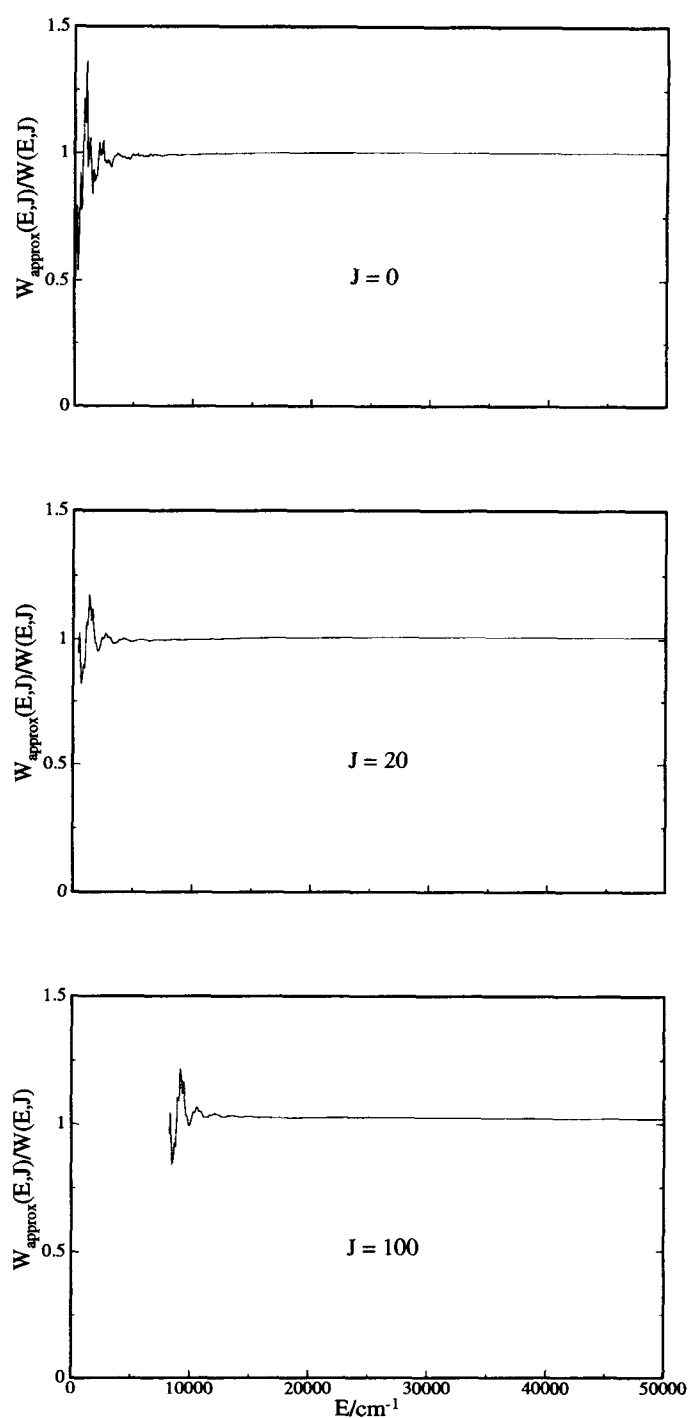


Fig. 7
As Fig. 6, but for CH_3OH . The deviations immediately above the starting point of the curves are much smaller than in the case of H_2O_2

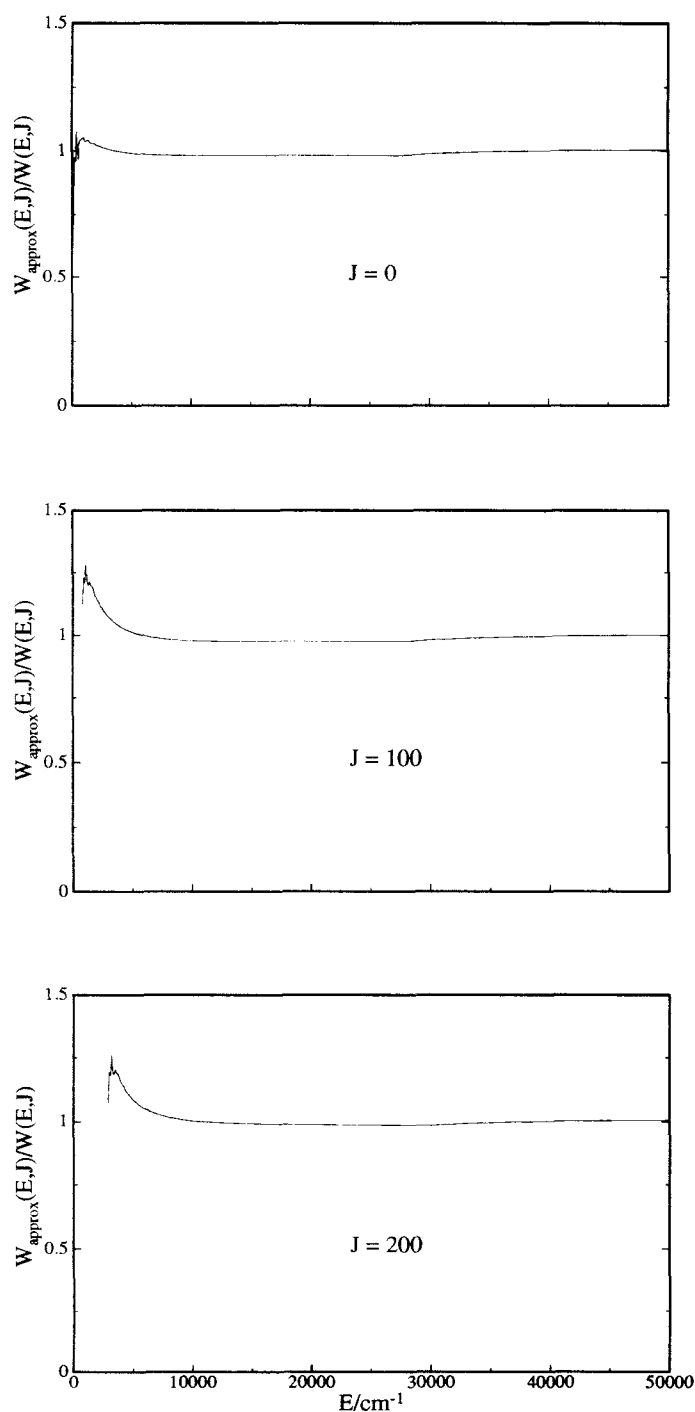


Fig. 8
As Fig. 6, but for C_7H_8

used for a quick calculation of $W^\ddagger(E, J)$. However, while the threshold energies $E_0(J)$ are usually sufficiently large for the Whitten-Rabinovitch results of Section 2 about $\rho(E, J)$ to apply, this is not necessarily the case, if $W^\ddagger(E, J)$ is calculated with the results from Section 3, i.e. if

$$W^\ddagger(E, J) = W_{WR}(E^*)F_W(E^*, J) \quad (4.1)$$

with F_W from Eq. (3.5) and

$$E^* = E - E_0(J=0) - B^\ddagger J(J+1) \quad (4.2)$$

is used ($r, s, A, B, C, \varepsilon_i, B_j$ now are adapted for the rigid activated complex). As often energies close to threshold, i.e. small E^* -values, are of interest, one leaves the range of good performance of the Whitten-Rabinovitch approximation. However, with some minor modifications, satisfactory results are obtainable: First, replacing $W_{WR}(E^*)$ by $\text{Int}(W_{WR}(E^*) + 1)$ in Eq. (4.1) particularly for larger molecules markedly improves the result [3] ($\text{Int}(x)$ denotes the largest integer below x). Replacing F_W from Eq. (3.5) by $\text{Int}(F_W)$ leads to a further improvement. The performance of the first modification was documented in [3]: the results were surprisingly close to the accurately counted numbers of states. (One should emphasize that adequate symmetry numbers have to be added at the end [1]).

5. Conclusions

The present work proposes approximate analytical expressions for harmonic rovibrational densities of states $\rho(E, J)$ for spherical top, prolate and oblate symmetrical top, and linear molecules. Likewise numbers of states $W(E, J)$ are approximated for prolate symmetric top molecules. The results are summarized in Table 5. They serve for quick calculations of the harmonic densities of states $\rho(E, J)$ in the specific rate constants $k(E, J)$ of unimolecular reactions. For rigid activated complex situations, the numbers of states $W^\ddagger(E, J)$ for prolate symmetric tops are calculated in a similar way. The derived expressions are useful when large numbers of specific rate constants, particularly J -dependences besides E -dependences are required. As long as the fine effects of anharmonicity in $\rho(E, J)$ and or rigidity in $W^\ddagger(E, J)$ cannot be characterized with sufficient certainty, the present very economic calculations provide results of a quality which is equivalent to that of exact counting techniques.

Table 5

Summary of approximate analytical expressions for rigid rotor-harmonic oscillator rovibrational densities of states a) and number of states b)

a) $\rho(E, J) = \rho(E^*)F_\rho(E^*, J)$

Character of symmetric top	E^*	x	F_ρ	Condition
Spherical	$E - BJ(J+1)$	—	$2J+1$	—
Prolate	$E - BJ(J+1)$	$\frac{(A-B)J^2}{E^* + a(E^*)E_z}$	$\left(\frac{J^2}{x}\right)^{1/2} \frac{\Gamma(\frac{1}{2})\Gamma(s)}{\Gamma(s+1/2)} \left\{ \tanh \left[\frac{\Gamma(s+1/2)}{\Gamma(s)\Gamma(1/2)} \left(1 + \frac{2+s}{10}x\right) \frac{x^{1/2}}{0.5} \right] \right\} + 1$	$x \geq 0$
Oblate	$E - BJ(J+1) + (B-C)J^2$	$\frac{(B-C)J^2}{E^* + a(E^*)E_z}$	$\left(\frac{J^2}{x}\right)^{1/2} 2x^{1/2}(1-x)^{s-1} \left[\sum_{i=0}^{s-1} \frac{\Gamma(s)}{\Gamma(2i+1)\Gamma(i+1)\Gamma(s-i)} \left(\frac{x}{1-x}\right)^i \right] + 1$ $\left(\frac{J^2}{x}\right)^{1/2} \frac{1}{x^{1/2}} \left[\frac{s}{s^2 - 1/4} + \frac{1}{(2s^2 - 1/2)x} \right]$	$1 > x \geq 0$ $x \geq 1$
Linear Triatomic	$E - BJ(J+1)$	$\frac{\varepsilon_b(J+1)}{E^* + a(E^*)E_z}$	$3x + \left(\frac{2B(J+1)}{\varepsilon_b} - 3 \right) x^2 + \left(1 - \frac{3B(J+1)}{2\varepsilon_b} \right) x^3$ $\left[1 + \frac{B(J+1)}{2\varepsilon_b x} \right]$	$1 \geq x \geq 0$ $x \geq 1$

b) $W(E, J) = W_{\text{int}}(E^*)F_W(E^*, J)$

Character of symmetric top	E^*	x	F_W	Condition
Prolate	$E - BJ(J+1)$	$\left(\frac{(A-B)J^2}{E^* + a(E^*)E_z} \right)$	$\left(\frac{J^2}{x}\right)^{1/2} \frac{\Gamma(1/2)\Gamma(s+1)}{\Gamma(s+3/2)}$ $\times \left\{ \tanh \left[\frac{\Gamma(s+3/2)}{\Gamma(s+1)\Gamma(1/2)} \left(1 + \frac{3+s}{10}x\right) \frac{x^{1/2}}{0.5} \right] \right\} + 1$	$x > 0$

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